

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B. Tech CSE Cyber security**

Semester: VI

Name of the Course: Network and System Security

Course Code: TCS679

Time: 90 Minutes

Maximum Marks: 50

Note:

- i. Answer all the questions by choosing any one of the sub questions.
- ii. Each questions carries 10 marks.

Q1		
(a)	Explain the following general security requirements. Forward secrecy, backward secrecy and non-repudiation.	(10 Marks) CO1
	OR	
(b)	What is the use of a cryptosystem? What are the different component of a cryptosystem?	
Q2		
(a)	How would you compare the symmetric and asymmetric encryption techniques?	(10 Marks) CO3
	OR	
(b)	Demonstrate the scenario of a denial-of-service (DoS) attack using a figure.	
Q3		
(a)	What are the uses of the following security mechanisms? Traffic padding, notarization and routing control.	(10 Marks) CO2
	OR	
(b)	Encrypt the plaintext "HELP" using the Hill Cipher with the following 2×2 key matrix : $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$ Assume the standard A=0, B=1, C=2, ..., Z=25 mapping for letters.	
Q4		
(a)	Encrypt the message M = 9 using the RSA encryption scheme with the following values: Prime numbers p= 3, q=11 and public exponent e=7	(10 Marks) CO3
	OR	
(b)	How would you compare cipher feedback mode and output feedback mode of operations of block ciphers?	
Q5		
(a)	Explain the concept of authentication tied to plaintext in cryptographic security using a figure. How does it ensure message integrity and authenticity without relying on encryption?	(10 Marks) CO5
	OR	

(b)	Differentiate between unconditionally secure and computationally secure in cryptographic systems. Why is computationally secure more commonly used in modern encryption schemes such as RSA?	
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